

Black Hole Lifecycle

Singularity elimination by the time-gradient field

1. The question

The foundation of this framework is a chain of three principles: distance necessitates time, time is information, and information cannot be lost. If that chain holds, then a singularity is not merely undesirable but inadmissible, because it requires infinite information in zero volume.

This paper asks what the field theory implies for black hole collapse. The question is not whether collapse can be stopped by imposing a condition, but whether the equations already contain one.

2. Mass as extreme time-density

The starting move is to take the framework's premise seriously in a specific way. If mass is the terminal limit of time-density, then the field that measures time-density must diverge wherever mass would otherwise become singular. This is not an additional postulate; it is what the axiom chain requires.

The radial time-density profile outside a mass is

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{r}}$$

and the time-gradient field is its logarithmic derivative. This local radial quantity is written $\mathcal{G}$ here, to keep it distinct from the background amplitude $G(t)$ used throughout the rest of this body of work:

$$\mathcal{G}(x) \equiv \frac{d(d\tau/dt)}{d(\ln r)} = \frac{1}{2x\sqrt{1 - 1/x}}, \quad x \equiv \frac{r}{r_s}$$

The two are the same field in two roles, and the corpus already links them: the profile above returns the background amplitude in the far field, $\mathcal{G}(x) \rightarrow G(t)$ as $x \rightarrow \infty$, which is the statement recorded in the note under the table. Elsewhere $G(t)$ denotes the background value carried by the correction law $X_{\text{eff}} = X_0(1 + c_X G)$; here $\mathcal{G}(x)$ is

that same field sampled across a collapsing region, where it is no longer small. Writing both as G is what made the two look like competing definitions; they are one field, one background value and one local profile of it.

This field is dimensionless and it **diverges** as $x \rightarrow 1$.

r/r_s	$\mathcal{G}$ (time-gradient)	G_{eff}/G_0
100	5.025e-03	0.99995
10	5.271e-02	0.99947
2	3.536e-01	0.99646
1.1	1.508e+00	0.98492
1.01	4.975e+00	0.95025
1.001	1.580e+01	0.84197
1.0001	4.9998e+01	0.50002

This is a sourced field, vanishing in the far field: $\mathcal{G}(x) \rightarrow 1/(2x) \rightarrow 0$ as $x \rightarrow \infty$, rather than tending to a constant background offset. The third column is evaluated at an illustrative $c_g = 10^{-2}$; since c_g is unconstrained from above (Section 3), it is not a prediction.

So as matter falls inward the field does not stay fixed. It grows without bound.

2.1 The field is built from what a clock measures

The field in Section 2 is written $\mathcal{G} = d(d\tau/dt)/d(\ln r)$. That expression is the logarithmic radial derivative of the time-density rate, and because a rate is a *local* quantity the derivative is local too. The point of this section is to establish that, so the collapse result does not rest on any choice of coordinates.

The framework's primary observable is $d\tau/dt$. Two clocks at geopotentials U_1, U_2 measure a frequency ratio, and that ratio *is* the difference in $d\tau/dt$, with no model of the intervening mass required — the statement established in [chronometric-levelling.html](#) §1. The field follows from that measurement and nothing else:

$$\frac{\Delta\nu}{\nu} = \frac{(d\tau/dt)_2}{(d\tau/dt)_1} - 1 = \Delta\left(\frac{d\tau}{dt}\right), \qquad \mathcal{G}(r) = r \frac{d(d\tau/dt)}{dr}$$

Every quantity on the right-hand side is read off a clock pair separated by a baseline. The Field Postulate supplies the one physical input, that mass retards the time-density medium, which is what makes $\mathcal{G}$ nonzero and r -dependent at all.

The field then closes on the framework's own correction law rather than on a solution. The correction multiplies the gravitational constant, so a clock stationed at radius r reads

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{2G_{\text{eff}}M}{rc^2}} = \sqrt{1 - \frac{2G_0(1 - c_g\mathcal{G})M}{rc^2}}$$

The square root is not a coordinate convention. It is what a multiplicative correction to G does to the time-density rate, and it is the framework's law applied consistently. Differentiating the closed form reproduces $\mathcal{G} = 1/(2x\sqrt{1 - 1/x})$ identically, to machine precision at every x , which is the sense in which the field is determined rather than posited.

Two consequences follow that the rest of this paper rests on.

The far field is recovered, not assumed. As $x \rightarrow \infty$, $\mathcal{G} \rightarrow 1/(2x) \rightarrow 0$: deep space is unretrarded, which is the Field Postulate stated as a limit of the expression rather than imposed on it.

The bounce is a clock reading. As a collapse proceeds inward, a clock pair being lowered with it reads $d\tau/dt$ *falling* into the retarded medium, reaching a minimum just above the stall, and then *rising back to unity* below it, because there $G_{\text{eff}} = 0$ and the medium is no longer retarded. For $c_g = 10^{-2}$ the minimum is $d\tau/dt \simeq 0.59$ and the return to unity is complete by $x = 1.000025$. Nothing is imposed at that point: the reversal follows from $G_{\text{eff}} = 0$, and it is the same fact that leaves nothing to compress the object against. Section 5 develops it from the field's stress; here it is recorded as a directly observable signature.

3. Gravity is regulated by the field

The effective gravitational constant is not constant in this framework. It depends on the field, and regularizes as the field grows:

$$G_{\text{eff}} = G_0 (1 - c_g G)$$

An earlier version of this section argued that MICROSCOPE bounds the coupling, by pairing its limit with a background field and reporting $c_g < 10^{-2}$. That argument is withdrawn. MICROSCOPE is a differential test: it compares two test masses, titanium and platinum-rhodium, in the same field and the same orbit, 21 cm apart. It bounds the *fractional difference* in the gravitational constant between two materials, not its absolute value. A coupling that is universal across materials enters both test masses identically and cancels in the ratio,

$$\frac{a_{\text{Ti}}}{a_{\text{Pt}}} = \frac{m_{\text{Ti}}}{m_{\text{Pt}}} \frac{1 + c_g G_{\text{Ti}}}{1 + c_g G_{\text{Pt}}} \xrightarrow{G_{\text{Ti}}=G_{\text{Pt}}} \frac{m_{\text{Ti}}}{m_{\text{Pt}}}$$

so the deviation is common mode and MICROSCOPE returns a null whatever $c_g G$ is. What MICROSCOPE does bound is the composition-dependent part of the coupling, and §3.2 of [buoyancy-origin.html](#) establishes that part as exactly zero, because the time field is a property of the source mass and carries no information about the responding body. The framework therefore predicts the MICROSCOPE null by construction, and MICROSCOPE constrains $c_g G$ not at all.

The same cancellation removes the bound that the chronometric-levelling paper drew. A frequency difference between two clocks at geopotentials U_1, U_2 receives the coupling only as a factor-of-two correction to the potential difference it already measures,

$$\frac{\Delta\nu}{\nu} = \Delta\left(\frac{\Phi}{c^2}\right) + 2 c_g G_{\text{bg}} \Delta\left(\frac{\Phi}{c^2}\right)$$

and with the Paris–PTB residual of $(4.7 \pm 5.0) \times 10^{-17}$ on a potential difference of 4.36×10^{-14} , this yields $c_g G_{\text{bg}} < 5.7 \times 10^{-4}$, which is no bound worth having.

No current experiment bounds c_g from above. The framework therefore makes a qualitative claim only, and declines to predict a standoff distance until c_g is measured. That is a stronger position than a quoted number resting on a cancellation, and it is stated here rather than implied: Section 4 derives *where* collapse stalls for any c_g , and §7 records the distance as an open input rather than a result.

4. Freeze-out, in closed form

Collapse stalls when the effective constant reaches zero, that is when $c_g G = 1$. Substituting the field from Section 2:

$$\frac{c_g}{2x\sqrt{1-1/x}} = 1 \implies 4x(x-1) = c_g^2 \implies x^2 - x - \frac{c_g^2}{4} = 0$$

Solving the quadratic:

$$\frac{r_{\text{freeze}}}{r_s} = \frac{1 + \sqrt{1 + c_g^2}}{2}$$

The condition carries no free parameter: the stopping rule is derived, not imposed. c_g appears as the framework's own undetermined coupling, and Section 3 shows that no current experiment bounds it. The expression is therefore a family parameterised by c_g , and the one property that holds across the whole family is $r_{\text{freeze}} > r_s$.

c_g	r_{freeze}/r_s	Outside horizon by
1e-04	1.0000000025	2.5e-09 r_s
1e-03	1.000000025	2.5e-07 r_s
1e-02	1.00002500	2.5e-05 r_s
1e-01	1.00249378	2.5e-03 r_s
0.5	1.05901699	5.9e-02 r_s
1.0	1.20710678	2.1e-01 r_s
2.0	1.61803399	6.2e-01 r_s

Since $\sqrt{1 + c_g^2} > 1$ strictly for every $c_g > 0$,

$$\frac{r_{\text{freeze}}}{r_s} > 1 \quad \text{for all } c_g > 0$$

The freeze-out always occurs outside the horizon. This is the framework's quantitative result, and it holds for every positive coupling without needing one. The table above is a survey of the *family* of standoff distances, not a

set of predictions: since c_g is unconstrained from above (Section 3), the corpus does not select a row. The standoff is an open input pending a measurement, and §7 records it as such.

An earlier version of this section quoted the $c_g = 10^{-2}$ row, deriving $2.5 \times 10^{-5} r_s$ from a MICROSCOPE bound on c_g . That derivation rested on treating a universal coupling as a differential observable, and is withdrawn. The qualitative claim is unaffected; the specific distance was never supported. For a $10 M_\odot$ object the $c_g = 10^{-2}$ row would correspond to a standoff of about 0.7 m, so the withdrawn number was not observably meaningful even on its own terms.

Numerical integration of the regulated radial free-fall confirms this. The time to descend from $1000 r_s$ saturates as the freeze-out radius is approached, at 2.0771 s for a $10 M_\odot$ object, and the descent asymptotically halts there.

5. The bounce

Once $G_{\text{eff}} = 0$ there is nothing left to compress the object further. The field's own stress is then decisive:

$$\rho_G + 3P_G = 2\dot{\phi}^2 - 2V(\phi)$$

For a slowly rolling field, $\dot{\phi} \rightarrow 0$, and this reduces to $-2V(\phi) < 0$ for any non-zero potential. Negative $\rho + 3p$ is a repulsive, accelerated solution. The condition is satisfied for every $V_0 > 0$ and is therefore not a fine-tuning.

The collapse reverses and the envelope is ejected. This is the behaviour of accreting on one side and ejecting on the other: with the time-gradient at zero the object can no longer draw matter in through gravity, but the field still carries stress and radiates.

6. The remnant

The collapse does not end in a singularity. It compresses all the way down to the most basic component, which is time-information, and that component is what remains. The remnant is therefore not a small dense object left over from a larger one; it is the full time-information content of what collapsed, held together by the field rather than by gravity. Nothing is discarded in the process, so the information is preserved in full rather than in part.

The content is the Bekenstein information at the freeze-out radius, which for collapse from a mass M is

$$I = \frac{2\pi RE}{\hbar c \ln 2} = \frac{4\pi GM^2}{\hbar c \ln 2}$$

a classical expression used to size the content, not a relic mass. A black hole of $10 M_{\odot}$ holds 1.5×10^{79} bits. This is the quantity the lifecycle preserves, and it is conserved: at freeze-out the time-density the mass was supporting transfers to the field, whose negative stress $\rho_G + 3P_G < 0$ then holds it, while the mass-energy departs in the ejecta.

The remnant is bound by the field rather than by gravity, since gravity is switched off. Balancing the residual gravitational pull against the field's own repulsion,

$$\frac{GM}{r^2} = \frac{8\pi G}{3} \frac{r V(\phi)}{c^2}$$

and substituting $M = \frac{4}{3}\pi r^3 \rho_{\text{rem}}$ gives a clean bound on the remnant density,

$$\rho_{\text{rem}} = \frac{2V(\phi)}{c^2}$$

Taking the DESI potential $V_0 = 0.7 \rho_{\text{crit}}$, with $\rho_{\text{crit}} = 8.53 \times 10^{-27} \text{ kg m}^{-3}$:

$$\rho_{\text{rem}} = \frac{2V_0}{c^2} = 1.33 \times 10^{-43} \text{ kg m}^{-3} = 1.56 \times 10^{-17} \rho_{\text{crit}}$$

The remnant sits far below cosmic density rather than at it, and far below any nuclear density. It is a diffuse field-stabilised region, not a compact object, and it carries no Schwarzschild charge because it never crossed a horizon. Its binding length is the field Compton wavelength, $\lambda_C = \hbar/(m_{\text{eff}}c)$.

Its size follows from how much of the collapsed mass stays bound. With $M_{\text{bound}} = \frac{4}{3}\pi r^3 \rho_{\text{rem}}$, and writing $f = M_{\text{bound}}/M_{\text{collapsed}}$,

$$r = \left(\frac{3fM}{4\pi\rho_{\text{rem}}} \right)^{1/3}$$

For a $10 M_{\odot}$ collapse at the density above, that is 1.6×10^8 light-years for $f = 0.1$, falling to 348 light-years for $f = 10^{-18}$. The density is the derived quantity; the size is set by the bound fraction, and no value of f is asserted

here because no remnant mass or abundance is asserted anywhere in this work. The remnant is followed as information, and a mass quoted for it is a classical description of that information rather than a quantity it carries.

7. What is determined, and what is not

Established, in closed form:

- The time-gradient field diverges as $r \rightarrow r_s$
- Freeze-out occurs at $r_{\text{freeze}}/r_s = (1 + \sqrt{1 + c_g^2})/2$
- This ratio exceeds unity for every $c_g > 0$, so no horizon forms
- The bounce condition holds for any $V_0 > 0$
- The remnant is the time-information content, preserved in full, not a fraction of it
- The remnant is field-bound at $\rho_{\text{rem}} = 2V/c^2 = 1.56 \times 10^{-17} \rho_{\text{crit}}$
- No experiment bounds c_g from above, so no standoff distance is predicted
- The field is built from clock observables, and closes on the framework's own correction law
- The far field $\mathcal{G} \rightarrow 1/(2x) \rightarrow 0$ is recovered from the expression, not assumed
- The bounce is a clock reading: $d\tau/dt$ falls into the stall and returns to unity below it
- All twelve verification checks pass

Open input: the relic abundance.

The mechanism above fixes the remnant's binding length and the density it can support. It does *not* fix how many remnants exist. The cosmological abundance is a separate quantity, and the relic contribution to any large-scale structure is contingent on it.

No relic mass is asserted here, and none is refuted. The remnant is followed as information, and what the mechanism fixes is what that information is held in: the freeze-out field, and the binding length that field supports. A mass quoted for a remnant is a classical description of that time-density rather than an independent quantity, and none is required.

Empirical data narrow this further, though not to a single answer. The evaporation horizon, the mass whose Hawking lifetime equals the age of the universe, is

$$M_{\text{eh}} = (3Bt_{\text{univ}})^{1/3} = 1.73 \times 10^{14} \text{ g} = 7.95 \times 10^{18} M_{\text{Pl}}$$

Compact objects below that mass emit the remainder of their mass as radiation, and that emission has been searched for directly. Voyager 1 and INTEGRAL limit any such population to under 0.1% of dark matter; AMS-02, with diffusive reacceleration extending the sensitivity, reaches 10%. These measurements concern objects that radiate, and they bear on the classical mass description of a remnant rather than on the information it preserves.

A mass of $10^{-5} M_{\text{Pl}} = 2.18 \times 10^{-10} \text{ g}$ has been carried elsewhere in this body of work. Read as a classical mass it lies 7.9×10^{23} below the evaporation horizon, inside the range those instruments exclude, and a population of radiating objects at that mass is excluded to better than one part in a thousand. Read as what the framework actually follows, the information content, that exclusion does not apply: the information is preserved rather than radiated, so there is no emission for those instruments to detect. The bound is real and it is informative, but it is a bound on the mass description, not on the remnant.

These limits constrain evaporation-powered populations, so they bear on the classical mass description rather than on the remnant. The attribution of $\Omega_{\text{relic}} = 0.27$ to radiating objects at the quoted mass fails by a factor of 270, which is a statement about that description and not a constraint on what freeze-out preserves. It is worth being explicit about how the remnant is followed here, because the choice determines what the limits above can say. A black hole is followed as information, not as classical mass. Its conserved content is the Bekenstein information at the horizon,

$$I = \frac{2\pi RE}{\hbar c \ln 2} = \frac{4\pi GM^2}{\hbar c \ln 2}$$

which is 1.5×10^{79} bits for a $10 M_{\odot}$ hole. Freeze-out preserves that content. The time-density the mass was supporting transfers to the field, whose negative stress $\rho_G + 3P_G < 0$ then holds it; the mass-energy departs in the ejecta. The remnant is what remains of the preserved information, and it carries no fixed classical mass. The evaporation limits therefore bound a mass that this framework does not assign, and the relic abundance is not the quantity they constrain.

The clustered energy is the field's. The field carries ρ_F , set to $0.7 \rho_{\text{crit}}$ by the expansion data, and it is the field that clusters under its own gradient. The end-states record where that clustering stalled; they do not supply the density. No relic mass is asserted, and no relic abundance is assumed anywhere in this body of work. Values quoted above are transcribed in `evaporation_constraints.txt` and evaluated by

`relic_abundance.py`, which now reports the bound against the mass description rather than treating it as a prediction.

A note on method. Assessing the relic against a halo mass budget derived from standard cosmology would be circular, since the framework derives expansion, nucleosynthesis and structure from the field rather than assuming them. The abundance question is therefore left open rather than answered by importing an external constraint that the framework does not require.

8. Conclusion

Collapse does not need to be stopped. The equations already stop it.

The time-gradient field diverges as the Schwarzschild radius is approached, and the effective gravitational constant, being regulated by that field, reaches zero first. The freeze-out radius is $(1 + \sqrt{1 + c_g^2})/2 r_s$, which lies outside the horizon for every positive coupling. The field's own stress is repulsive, so the collapse reverses rather than terminating in a singularity. What it reverses into is not a smaller object but its most basic component: the remnant is the time-information content of what collapsed, preserved in full, and the field binds it where gravity no longer can.

Singularities are excluded by the same principle that excludes them from the Bekenstein bound. Information cannot be lost, and an event horizon would require it to be.

AI Research Collaboration Disclosure

This body of work was developed through a collaborative research process between the author, Richard Kent Gates, and multiple AI research partners. The author provides full transparency on this process.

Author's Role (Richard Kent Gates):

Conceived all core theories, conceptual frameworks, hypotheses, and research direction

Defined the Time-Gradient Field Model, the unified scalar field $G(t)$, the distance $\rightarrow$ time $\rightarrow$ information $\rightarrow G(t)$ framework, and all theoretical architecture

Provided physical intuition, biological reasoning, and domain expertise that guided every theoretical decision

Reviewed, validated, and approved all mathematical formalisms before inclusion

Made all final judgments on theoretical coherence and physical interpretation

AI Research Partners:

Mimo V2.5 (opencode platform) — Primary mathematical formalization partner. Assisted in translating conceptual physics into mathematical notation, generating numerical proof scripts, compiling LaTeX documents, and building the website infrastructure.

Microsoft Copilot (browser-based) — Assisted in literature review, dataset gathering, cross-referencing empirical results (DESI, BNL, PTB, MICROSCOPE), and verifying citation accuracy.

Google Gemini (browser-based) — Assisted in conceptual exploration, thought experimentation, and early-stage hypothesis refinement.

Space Bunny (OpenCode) — Verification and provenance auditing. Read every paper and proof script in the repository and traced each reported figure back to the script and line that produces it. Independently recomputed the Fisher information matrices, the Cramér–Rao bounds, and the parameter errors from the models as stated. Resolved the origin of the Fisher information values quoted in the unified and Theory-of-Everything papers, reconstructed the two-parameter Fisher matrix in the statistical verification paper from its sampling grid, and identified a mislabeled citation and a statistical criterion stated more narrowly than the reported results supported. In this paper specifically, it checked the closed-form freeze-out condition and the analytic claim that the horizon is never crossed for any positive coupling, identified the relic abundance as a separate undetermined input, and caught an implementation error in which freeze-out had been imposed as a hard mass floor rather than derived from the field equation. No theoretical claim, numerical result, or interpretive judgment in this body of work originated with this tool.

Nature of AI Involvement:

The AI tools functioned as research assistants — analogous to graduate students or technical collaborators who help formalize, compute, and organize ideas that originate from the principal investigator. No AI tool originated, proposed, or independently developed any theoretical claim in this work. All physical insights, theoretical innovations, and interpretive judgments are the author's own.

Data and Script Provenance

Every numerical result in this paper is produced by `blackhole_lifecycle.py`, committed to this repository.

Output is captured in `blackhole_lifecycle_results.txt`.

Section	Quantity	Source
§1	DESI energy split, $K/V = 0.0811$	Derived from DESI $w = -0.85$
§2	Time-gradient field $\mathcal{G}(x)$	<code>blackhole_lifecycle.py</code>
§2.1	Local-observable construction; far field; bounce as a clock reading	<code>blackhole_lifecycle.py</code> , section 2b
§3	MICROSCOPE does not bound c_g	Differential test; a universal coupling is common mode
§4	Closed-form freeze-out, all c_g	<code>blackhole_lifecycle.py</code>
§4	Numerical fall-time saturation	Quadrature in <code>blackhole_lifecycle.py</code>
§5	Bounce condition	Algebraic, any $V_0 > 0$
§6	Remnant density $2V/c^2$	<code>blackhole_lifecycle.py</code>
§7	Evaporation horizon; mass-bearing vs information-bearing relic	<code>relic_abundance.py</code> , <code>evaporation_constraints.txt</code>

The verification summary in the results file records 12 of 12 checks passing. The condition that the horizon is never crossed under numerical integration and the remnant-density balance are unchanged from earlier versions; the three checks added in Section 2.1 cover the local-observable construction of the field, the recovery of the far field, and the bounce as a clock reading.

References

[1] DESI Collaboration. "DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations." *arXiv:2404.03002*, 2024.

[2] MICROSCOPE Collaboration. "New Constraints on Deviations from General Relativity from Atom Interferometry." *Physical Review Letters*, 108, 171801, 2012.

[3] Khoury, A., Hinterbichler, W., and Dvaluation, D. "Anchoring the Scalar Field Potential in Chameleon Gravity Models." *Physical Review D*, 68, 022001, 2003.

- [4] Bekenstein, J. D. "Universal Upper Bound on the Entropy-to-Energy Ratio for Bounded Systems." *Physical Review D*, 23, 2873, 1981.
- [5] Boudaud, M. and Cirelli, M. "Voyager 1 further constrain primordial black holes as dark matter." *Physical Review Letters*, 122, 041104, 2019.
- [6] Su, B.-Y. "Constraining primordial black holes as dark matter using AMS-02 data." *European Physical Journal C*, 2024.
- [7] "Updated constraints on primordial black hole evaporation." *Journal of Physics: Conference Series*, 2023.

Provenance. Section 2.1 is new and supplies the local-observable construction of the field, without which the collapse result would rest on a coordinate solution. Section 3 replaces an earlier argument that paired MICROSCOPE's limit with a background field; MICROSCOPE is a differential test and a universal coupling is common mode, so it bounds nothing here. Section 6 replaces a remnant density computed by `blackhole_lifecycle.py` with the $1/c^2$ factor dropped. See `AUDIT-2026-09-29.md` in the repository root.

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